A harmonized trial-level dataset of picture-word interference

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Abstract

Decades of picture-word interference (PWI) research have produced a rich literature on how distractor words modulate picture naming, yet the trial-level data underlying these findings remain fragmented across laboratories, repositories, and bespoke file formats, limiting large-scale reuse and cross-study comparison. Here we present a harmonized, openly accessible dataset comprising 688,976 trials from 86 experiments across 42 studies, involving 3,353 participants naming pictures in English or German. Data were assembled from five sources: an in-house laboratory archive, raw data from a prior meta-analysis, systematic literature searches (OpenAlex, PubMed), direct author contributions, and the Open Science Framework. All datasets were cleaned and harmonized into a common 28-variable schema covering trial identifiers, response times, a harmonized response-accuracy code, experimental design characteristics, target–distractor relatedness indicators, and basic psycholinguistic properties of target and distractor words. Technical validation shows the expected reaction-time distributions across studies. The full data-collection and cleaning pipeline is openly released so that the resource is *extensible*: future PWI studies, additional languages, and further harmonized variables can be added by other researchers without rebuilding the infrastructure from scratch. The dataset enables mega-analyses, computational modeling of naming latencies, replication work, and methodological investigations that require statistical power beyond what individual studies can provide.

Keywords: picture-word interference; language production; lexical access; open data; mega-analysis; response times

1 Background & Summary

The picture-word interference (PWI) paradigm is one of the most widely used experimental tools for investigating lexical access during language production (Bürki et al., 2020; Cutting & Ferreira, 1999; Glaser & Düngelhoff, 1984; Korko et al., 2024; Lupker, 1979; Miozzo & Caramazza, 2003). In a typical PWI task, participants name a picture (the target) while ignoring a written or spoken distractor word simultaneously or with a small lag (Abdel Rahman & Aristei, 2010; Bürki & Madec, 2022). The key finding is that naming latencies tend to be modulated by the relationship between target and distractor words (Cutting & Ferreira, 1999; Starreveld & La Heij, 1996).

Over the past four decades, PWI research has generated a rich body of findings. The paradigm has been used to study mechanisms of lexical selection (Abdel Rahman & Aristei, 2010; Lupker, 1979), the role of inhibitory control in production (Broos et al., 2018), the time course of semantic and phonological encoding (Cutting & Ferreira, 1999; Sailor et al., 2009), and individual differences in naming performance (Ward et al., 2021). Recent meta-analyses have quantified effect sizes for semantic interference (Bürki et al., 2020) as well as for distractor frequency and distractor length on naming latencies (Bürki et al., 2023).

Despite this extensive literature, a critical gap remains: no harmonized, large-scale, trial-level PWI dataset exists that spans multiple studies, languages, and laboratories. Individual datasets are typically collected for specific theoretical questions and remain in study-specific formats, often with inconsistent variable naming, coding, and exclusion criteria.

The present dataset addresses this gap. We compiled trial-level data from 42 studies encompassing 86 experiments, 3,353 participants, and 688,976 trials in English and German. Data were sourced from five channels: (1) a large in-house dataset from the Neurocognitive Psychology Lab at *Humboldt-Universität zu Berlin*, (2) raw data from the meta-analysis by Bürki et al. (2020), (3) systematic literature searches via OpenAlex and PubMed, (4) direct author contributions in response to data-sharing requests, and (5) datasets identified through the Open Science Framework (OSF). All datasets were cleaned and harmonized into a common schema of 28 variables that capture the trial-level response (e.g. response time, accuracy), experimental design characteristics (e.g. stimulus onset asynchrony, collection setting), and basic psycholinguistic properties of target and distractor words (e.g. word length, word frequency).

This dataset serves multiple purposes. First, it enables mega-analyses that estimate effects across studies with far greater statistical power than any individual experiment. Second, the harmonized format facilitates reproducible replication studies and cross-study comparisons. Third, the trial-level structure supports computational modeling approaches. Fourth, the dataset provides a resource for methodological research on topics such as the impact of experimental design choices on PWI effects. Fifth, the dataset is designed to be *extensible*: the full pipeline (raw data, study-specific cleaning scripts, harmonization code, and the merged file) is deposited as a self-contained, openly licensed project, so that future studies can be incorporated by adding a single cleaning script and re-running the merge.

2 Methods

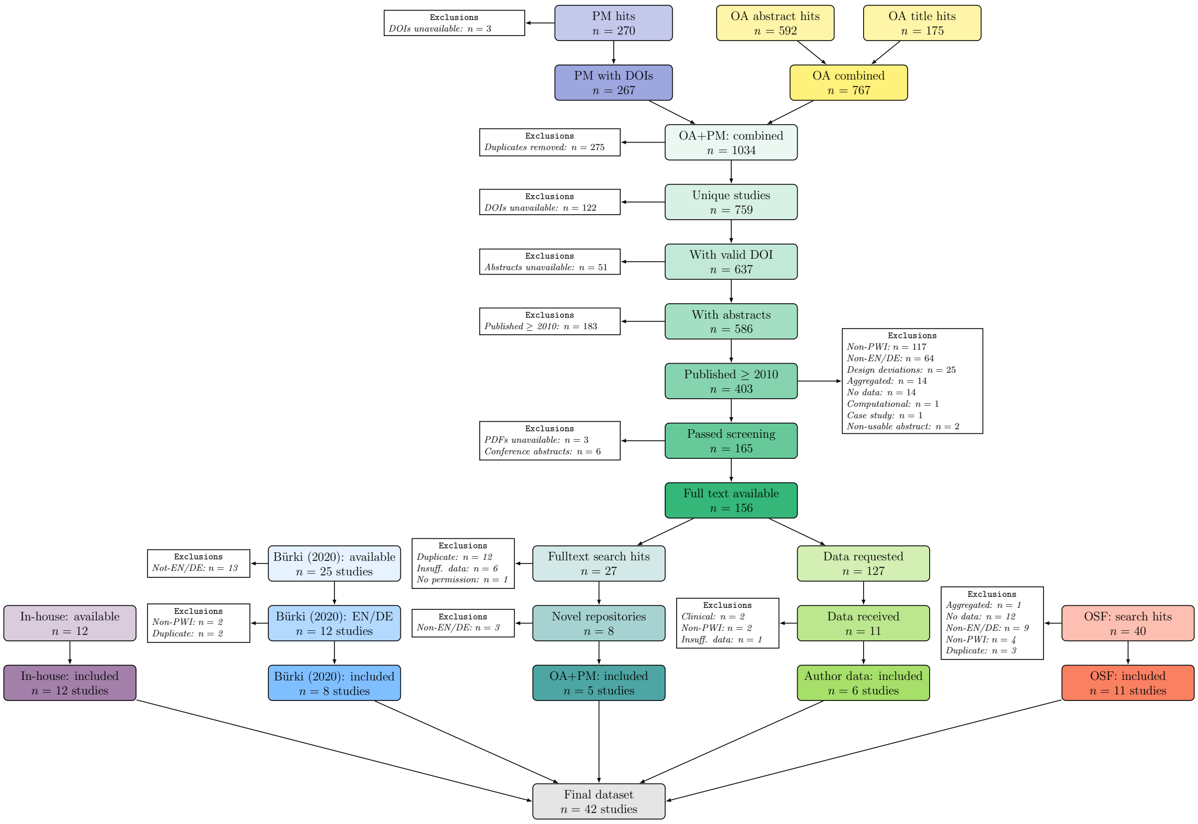
2.1 Data Collection

2.1.1 Overview of the Data Collection Process

Our objective was to compile a comprehensive, curated dataset of empirical studies employing the picture-word interference paradigm. This dataset was assembled through five complementary routes: (i) inclusion of a large in-house dataset collected by the Neurocognitive Psychology research group at Humboldt-Universität zu Berlin, (ii) incorporation of raw datasets from an existing meta-analysis by Bürki et al. (2020), (iii) systematic searches of the academic literature using OpenAlex and PubMed, (iv) direct requests for raw data from corresponding authors of eligible studies lacking publicly accessible data, and (v) targeted retrieval of publicly available data from the Open Science Framework (OSF). The integrated dataset comprises 86 experiments from 42 studies, totaling 3,353 participants and 688,976 trials. An overview of the data collection process is shown in Figure 1. Table 2 shows the total number of included studies, experiments, participants, and trials by data source, while Table 1 lists the individual studies, their data sources, and the number of participants and trials as well as summary statistics for their response times.

Figure 1

Overview of the data collection and screening process.



Note. OA = OpenAlex; PM = PubMed; Exclusion labels: Non-PWI = study does not use the classical picture-word interference paradigm; Non-EN/DE = non-English or non-German dataset; Insuff. data = insufficient trial-level data, i.e., data that do not satisfy our inclusion criteria (e.g., no target/distractor words available); No data = no trial-level data available; Duplicate = dataset already included via another source; Aggregated = only aggregated (not trial-level) data available; Clinical = clinical participant population; Design deviations = strong deviation from classical PWI paradigm (e.g., multiple distractors, delayed naming).

2.1.2 Inclusion and Exclusion Criteria

Studies were included if they met all of the following criteria:

- Used the classical picture-word interference paradigm (single target picture, single distractor word)
- Stimuli in English or German

- Non-clinical human adult participants
- Trial-level data available (individual response times per trial)
- Target and distractor words identifiable per trial

We included studies with both native and non-native speakers, as long as the materials were in English or German. Furthermore, studies were excluded if they did not conform strictly to the classical PWI paradigm. This included tasks such as Stroop, picture-word matching, delayed naming, sentence production, or experiments using multiple distractors per trial (Dhooge & Hartsuiker, 2011; Mädebach et al., 2022).

The restriction to English and German materials reflects two practical considerations. First, English and German are the two languages in which trial-level PWI data are most abundantly available, together accounting for the majority of published PWI experiments with shareable raw data; this makes them the natural starting point for a harmonized large-scale resource. Second, the authors are fluent in both languages, which was essential for validating stimuli (e.g. verifying categorical, associative, and phonological relations). Extending the dataset to additional languages is a natural direction for future releases (see Usage Notes).

2.1.3 In-House Data

We incorporated a large in-house dataset collected at the Neurocognitive Psychology Lab at Humboldt-Universität zu Berlin. All 12 studies fulfilled the inclusion criteria.

2.1.4 Meta-Analysis Data

We obtained raw datasets for 25 studies from a Bayesian meta-analysis by Bürki et al. (2020). We filtered the data to retain only studies conducted in English or German, yielding raw data for 28 experiments across 12 studies. Four studies were excluded at this stage: two due to substantial deviations from the classical PWI paradigm, and two due to duplication with a previously included study. Finally, we included 20 experiments across 8 studies from this data collection track.

2.1.5 OpenAlex and PubMed: Literature-Based Study Identification

2.1.5.1 Programmatic Data Collection. We performed systematic searches in January 2025 using the OpenAlex and PubMed databases. These searches were conducted using R (version 4.3.0). Through the `openalexR` package (Aria et al., 2024), two queries were executed on OpenAlex: one targeting abstracts (yielding 592 hits) and another targeting titles (175 hits). A parallel search using the `rentrez` package was conducted in PubMed (Winter, 2017), retrieving 270 records based on occurrences of the term 'picture-word interference' in titles or abstracts, of which 3 studies did not report valid DOIs and were excluded.

Table 1
Per-study summary of the 42 included studies.

Study ID	Citation	Language	Source	$n_{experiment}$	$n_{participant}$	n_{trials}	$M_{\mathbf{RT}}$	$SD_{\mathbf{RT}}$
aristei_2010	Abdel Rahman and Aristei (2010)	DE	Meta-analysis	1	30	1,161	754	219
abdel_rahman_unpub1	Abdel Rahman (n.d.-a)	DE	In-house	1	27	10,419	697	232
abdel_rahman_unpub2	Abdel Rahman (n.d.-d)	DE	In-house	1	31	19,740	827	261
abdel_rahman_unpub3	Abdel Rahman (n.d.-b)	DE	In-house	1	30	26,823	756	205
abdel_rahman_unpub4	Abdel Rahman (n.d.-c)	DE	In-house	1	22	10,250	983	293
aristei_2013	Aristei and Abdel Rahman (2013)	DE	Meta-analysis	1	30	11,351	822	223
broos_2018	Broos et al. (2018)	EN	OSF	2	176	24,092	799	327
burki_2022	Bürki and Madec (2022)	DE	OSF	1	45	15,275	902	272
catling_2010	Catling et al. (2010)	EN	Author request	1	16	1,437	1,014	332
cutting_1999	Cutting and Ferreira (1999)	EN	Meta-analysis	4	159	2,530	816	177
damian_2014	Damian and Spalek (2014)	DE	Meta-analysis	1	48	1,765	696	201
freund_2018	Freund and Nozari (2018)	EN	Repository	2	64	20,621	826	249
gauvin_2018	Gauvin et al. (2018)	EN	Meta-analysis	4	100	4,905	764	223
hantsch_2013	Hantsch and Mädebach (2013)	DE	Author request	1	29	9,845	680	157
hutson_2014	Hutson and Damian (2014)	EN	Meta-analysis	2	90	9,656	755	197
jescheniak_2020	Jescheniak et al. (2020)	DE	OSF	1	32	4,774	858	231
jescheniak_2024	Jescheniak et al. (2024)	DE	OSF	2	96	24,045	700	193
jescheniak_2024b	Jescheniak and Wöhner (2024)	DE	OSF	4	192	60,358	743	195

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Table 1

Per-study summary of the 42 included studies (continued).

Study ID	**Citation**	**Language**	**Source**	$n_{experiment}$	$n_{participant}$	n_{trials}	$M_{\mathbf{RT}}$	$SD_{\mathbf{RT}}$
kuhlen_2022	Kuhlen and Abdel Rahman (2022)	DE	In-house	3	96	14,917	820	248
kurtz_2018	Kurtz et al. (2018)	DE	OSF	3	96	26,079	679	173
lorenz_2018	Lorenz et al. (2018)	DE	In-house	1	60	38,077	783	243
lorenz_2019	Lorenz et al. (2019)	DE	In-house	1	64	40,672	915	283
lorenz_2021	Lorenz et al. (2021)	DE	In-house	1	31	4,933	810	217
lorenz_unpub1	Lorenz (n.d.-c)	DE	In-house	1	31	2,467	864	231
lorenz_unpub2	Lorenz (n.d.-b)	DE	In-house	1	31	10,189	843	248
lorenz_unpub3	Lorenz (n.d.-a)	DE	In-house	1	24	3,828	714	170
mascelloni_2021	Mascelloni et al. (2021)	EN	Repository	2	40	5,725	718	204
muylle_2024	Muylle and Jarema (2024)	EN	Repository	2	108	6,277	1,103	381
madebach_2018	Mädebach et al. (2018)	DE	Repository	1	72	11,824	775	222
madebach_2022	Mädebach et al. (2022)	DE	OSF	1	24	6,013	757	209
muller_2020	R. Müller (2020)	DE	Author request	3	77	18,056	849	350
sailor_2009	Sailor et al. (2009)	EN	Meta-analysis	6	144	8,437	777	226
spinelli_2019	Spinelli et al. (2019)	EN	OSF	1	48	6,333	905	324
vieth_2014a	Vieth et al. (2014a)	EN	Author request	2	104	34,345	675	231
vieth_2014b	Vieth et al. (2014b)	EN	Author request	3	104	19,322	690	182
vogt_2022	Vogt et al. (2022)	DE	In-house	3	144	22,343	1,060	306

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Table 1

Per-study summary of the 42 included studies (continued).

Study ID	**Citation**	**Language**	**Source**	$n_{experiment}$	$n_{participant}$	n_{trials}	$M_{\mathbf{RT}}$	$SD_{\mathbf{RT}}$
ward_2021	Ward et al. (2021)	EN	OSF	1	24	4,566	693	166
wei_2022	Wei et al. (2022)	EN	OSF	8	272	35,574	1,292	402
wei_2024	Wei et al. (2024)	EN	Repository	1	193	29,881	1,396	449
wohner_2022	Wöhner et al. (2022)	DE	OSF	6	298	75,697	829	314
de_zubicaray_2012	de Zubicaray et al. (2012)	EN	Author request	2	34	3,219	1,211	391
de_zubicaray_2013	de Zubicaray et al. (2013)	EN	Meta-analysis	1	17	1,155	800	170
Total				**86**	**3,353**	**688,976**	**853**	**331**

Note. M_{RT} and SD_{RT} are the mean and standard deviation of trial-level response times, pooled across all conditions, experiments, and participants of each study (RTs $\leq$ 150 ms or $>$ 3,000 ms and trials marked as technical errors were excluded). The **Total** row reports M_{RT} and SD_{RT} pooled across all 688,976 retained trials of the merged dataset, not the mean (or weighted mean) of the per-study values.

Table 2
Summary of included studies by data source.

Data source	$n_{studies}$	$n_{experiments}$	$n_{participants}$	n_{trials}
Author contributions	6	12	364	86,224
Bürki meta-analysis	8	20	618	40,960
Fulltext search hits (OpenAlex, PubMed)	5	8	477	74,328
In-house dataset (NKP)	12	16	591	204,658
OSF API	11	30	1,303	282,806
Total	**42**	**86**	**3,353**	**688,976**

Note. NKP: Neurocognitive Psychology Lab at Humboldt-Universität zu Berlin.

Combining results from both databases produced an initial dataset comprising 1,034 records. Removing duplicates based on OpenAlex IDs reduced the dataset to 759 unique records. Selecting only studies with DOIs reduced the dataset to 637 unique records. Language filtering using the `cld2` language detection package (Ooms, 2024) retained 627 English-language articles. Abstracts missing from this subset were retrieved from PubMed XML data using `entrez_fetch()`, resulting in 586 fully documented articles. Furthermore, to have a realistic chance of finding a dataset, we restricted the dataset to studies published from 2010 onward ($n = 403$).

2.1.5.2 Title and Abstract Screening. Each article's title and abstract was reviewed to confirm adherence to the PWI paradigm and to our inclusion and exclusion criteria. This screening process ultimately identified 165 eligible studies. Common reasons for exclusion included non-PWI tasks (117), non-English/German datasets (64), and deviations from the classical PWI design (25), followed by aggregated data (14), non-quantitative studies (14), computational-only studies (1), case studies (1), and missing abstracts (2).

2.1.5.3 Keyword-Based Search. From the 165 eligible studies, we gathered full-text PDFs wherever available. Three studies lacked accessible PDFs, and six were conference abstracts without accompanying data, leaving 156 full-text articles for further inspection. We systematically searched the remaining 156 articles for repository-related keywords, including 'OSF', 'Open Science Framework', 'osf.io', 'zenodo', 'figshare', and 'github'. This keyword-based screening yielded 27 hits. Of these, 12 datasets were already present in our collection, 5 were newly included, and the remaining 10 entries were excluded for the following reasons: 3 used non-English or non-German materials, 6 lacked sufficient or accessible data, and 1 required author permission for access, which was not granted by the end of the data collection process. We manually examined all remaining 129 articles and their supplementary files for additional data availability; however, no further datasets suitable for inclusion were identified. Thus, 5 datasets were newly added in this step.

2.1.6 Data Retrieval via Author Request

We contacted the corresponding authors of the remaining $n = 127$ eligible studies for which no repository link had been identified. These studies involved 84 unique authors. We received usable data for 11 studies, of which 5 were excluded for the following reasons: clinical populations (2), strong deviations from the classical PWI paradigm (2), and insufficient data (1). The remaining 6 studies met all inclusion criteria and were added to the final dataset.

2.1.7 OSF: Repository-Based API Data Retrieval

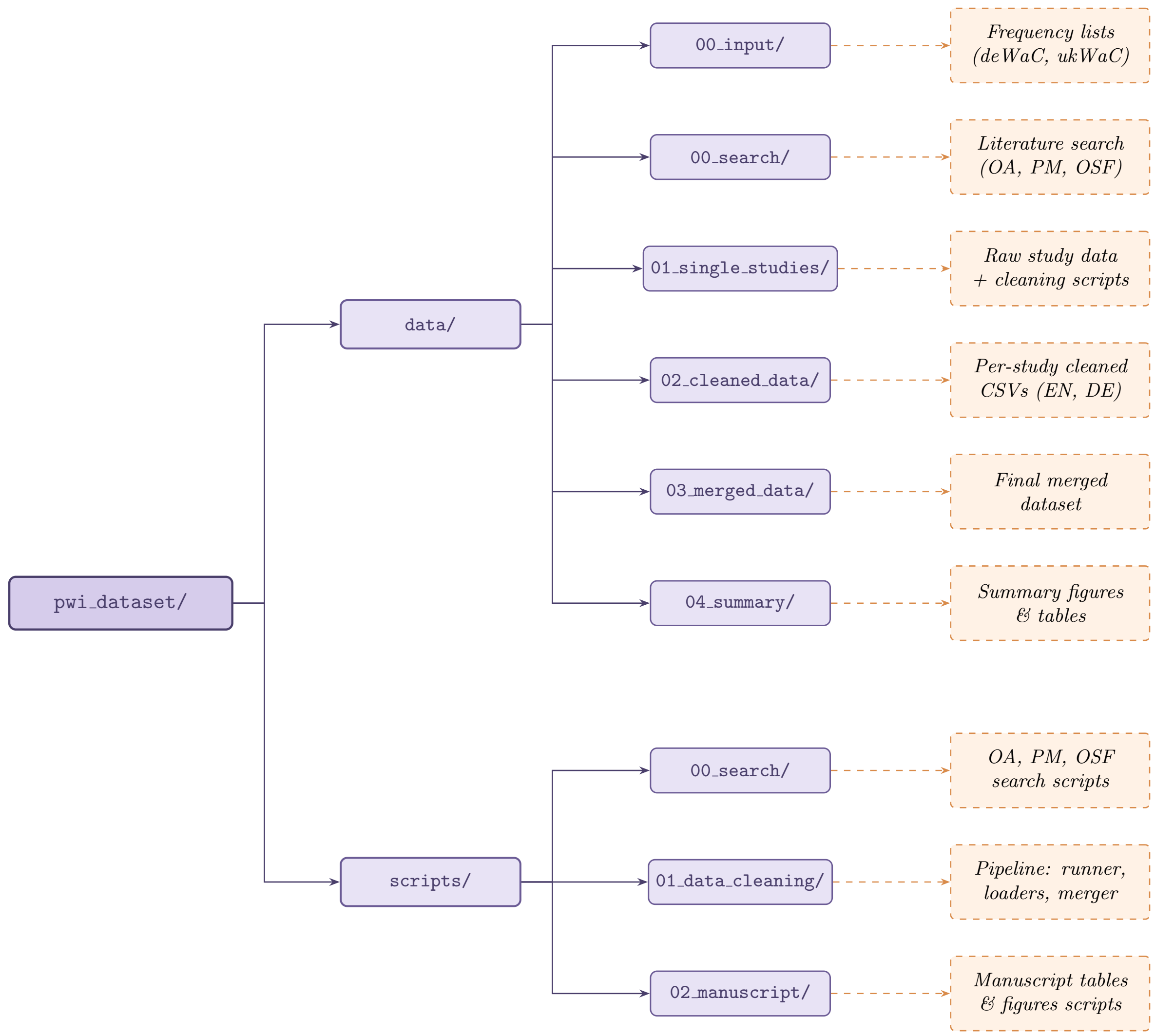
We queried the Open Science Framework (OSF) in January 2025 via its public API using the search term ("picture-word interference"). This search yielded 40 repositories, which were manually screened. Following this screening, 11 repositories were retained for inclusion, while 29 were excluded. Reasons for exclusion were aggregated or summary-level data (1), insufficient or missing data on OSF (12), non-English or non-German materials (9), non-conformity with the PWI paradigm upon closer inspection (4), and duplication with other data sources (3).

2.2 Ethics Statement

No new human-subjects data were collected for this work. All included studies had been approved by their respective institutional ethics committees as reported in the original publications.

Figure 2

Overview of the repository folder structure hosted on OSF.



Note. Top-level directories (`data/`, `scripts/`) group the raw inputs, processing pipeline, and manuscript-supporting outputs. Orange dashed boxes summarize the contents of each subfolder. PM = PubMed, OA = OpenAlex, OSF = Open Science Framework.

3 Data Records

We provide a fully reproducible pipeline that transforms raw study datasets into a single harmonized dataset. All materials including raw data, cleaning scripts, harmonization code, and the final merged CSV are openly available on the Open Science Framework (OSF) at https://doi.org/10.17605/OSF.IO/2B3SX. Of the 42 included studies, eight had already released their trial-level data under a CC-BY 4.0 license; for the remaining

34 studies, we obtained author permission to share the trial-level data under the same license.

The repository is organized around two top-level directories (`data/` and `scripts/`), whose numbered subfolders correspond to successive pipeline stages (Figure 2). Within `data/`, `00_input/` and `00_search/` hold inputs for frequency computation and literature-search outputs; `01_single_studies/` contains raw study data and study-specific cleaning scripts; `02_cleaned_data/` contains per-study cleaned CSVs (separated by language); `03_merged_data/` stores the final merged trial-level dataset; and `04_summary/` contains manuscript-supporting figures and summary tables. Scripts are analogously grouped under `scripts/00_search/`, `scripts/01_data_cleaning/`, and `scripts/02_manuscript/`. All scripts use the `here` package for portable, project-root-relative paths (K. Müller, 2025).

3.1 Harmonization

The harmonization pipeline proceeds in three sequential stages, each implemented by dedicated scripts in `scripts/01_data_cleaning/`.

3.1.1 Stage 1: Study-Specific Cleaning

Each source dataset is cleaned by a dedicated R script stored in `data/01_single_studies/<study>/Scripts/`. These scripts (i) harmonize per-study response codes into a common `accuracy` vocabulary with values `correct`, `wrong_word`, `other_error`, and `technical_error`, (ii) standardize variable names and coding conventions, and (iii) assign unique participant identifiers prefixed by the study name. Subject-side errors (e.g., wrong-word responses, disfluencies, non-speech voice-key triggers) are retained in the cleaned files so that users can apply their own exclusion criteria; only trials flagged as `technical_error` (voice-key failures, microphone artefacts) are dropped centrally at Stage 3. A runner script (`Stage1_execute_all_cleaning_scripts.R`) executes all study-specific scripts in isolated environments, reading raw data from `data/01_single_studies/<study>/Data/` and writing cleaned CSVs to `data/02_cleaned_data/<Language>/single_studies/`.

3.1.2 Stage 2: Language-Specific Loading and Frequency Computation

Two scripts (`Stage2_Read_cleaned_data_English.R` and `Stage2_Read_cleaned_data_German.R`) load the cleaned CSVs for each language and apply further preprocessing.

First, response times below 150 ms and above 3,000 ms are excluded. These conservative thresholds, attested in prior naming studies, remove implausibly fast responses and unusually slow latencies typically treated as outliers (Catling et al., 2010; de Zubicaray et al., 2012; Jescheniak et al., 2020; Mädebach et al., 2018).

Second, word frequency information is incorporated using the deWaC corpus for German and the ukWaC corpus for English (Baroni et al., 2009), and reported on the Zipf scale (Brysbaert et al., 2018).[1]

3.1.3 Stage 3: Merging

The merge script (`Stage3_Merge_dataset_paper.R`) combines the English and German datasets, stores binary variables as human-readable factor labels (e.g., `yes/no`, `text/audio`, `related/unrelated`) so the dataset can be inspected without an extra coding lookup, computes word length and repetition counts, excludes trials flagged as `technical_error`, and writes the final 28-column dataset to `data/03_merged_data/`.

Table 3 summarizes all variables, including their descriptions and value ranges. The 28 variables are organized into five thematic groups—identifiers, the trial-level response, experimental design characteristics, indicators of the target–distractor relationship, and psycholinguistic properties of target and distractor words. The remainder of this section describes each group in more detail, focusing on aspects that are not immediately apparent from the table.

3.1.3.1 Identifiers (variables 1–3). Each trial is uniquely indexed by `study_id`, `experiment_id`, and `participant_id`. Participant identifiers are anonymized and unique within the merged dataset, so no participant identifier is shared across studies even if the original raw data used overlapping numbering schemes.

[1]Zero-frequency items are handled with Laplace smoothing (adding one to each word's count and adjusting the total corpus size by the number of distinct lemmas; (Brysbaert & Diependaele, 2013)); Zipf frequencies are then computed as the base-10 logarithm of the smoothed frequency per billion words plus 3.

3.1.3.2 Trial-level response (variables 4–8). `trial_number` indexes the position of a trial within a participant's sequence and starts at 1 for each participant. `target_word` contains the picture name (i.e., the word the participant is expected to produce) and `distractor_word` the lexical distractor presented on that trial. `rt` is the response time in milliseconds: in overt-naming experiments it corresponds to the voice-onset latency, whereas in covert-naming experiments it corresponds to the latency of the manual response (see `task_type` below). `accuracy` encodes whether the response is `correct` (the target picture was named correctly), `wrong_word` (a different word than the target was produced, e.g., a synonym or a category coordinate), or `other_error` (any other subject-side error, including disfluencies, omissions, or non-speech sounds). Trials flagged as technical errors are removed centrally at Stage 3 and therefore do not appear in the merged dataset.

3.1.3.3 Experimental design (variables 9–17). `language` indicates the experimental language (`EN` for English, `DE` for German). `native_dutch` and `native_french` flag experiments in which participants were native speakers of Dutch or French rather than of the experimental language; these flags are included so that users can transparently include or exclude bilingual subsamples. `has_familiarization` indicates whether participants completed a familiarization phase (e.g., a pre-experimental picture-name training block) before the main task. `soa` is the stimulus onset asynchrony in milliseconds, defined as the temporal offset between distractor and target picture onset; negative values indicate that the distractor preceded the picture, zero indicates simultaneous onset, and positive values indicate that the picture preceded the distractor (Glaser & Düngelhoff, 1984). `task_type` distinguishes `overt` naming, in which participants produce the picture name aloud and reaction times are obtained from voice-onset detection, from `covert` naming, in which the spoken response is replaced by a manual response (typically a keypress) and reaction times are obtained from response-key onset (Wei et al., 2022). `has_additional_tasks` flags experiments that combined the PWI task with secondary tasks (e.g., a concurrent memory load), which can affect overall RT levels. `is_gamified` indicates whether the experiment was embedded in a game-like environment with interactive feedback, scoring, or narrative framing rather than a conventional laboratory routine (Wei et al., 2022, 2024). `collection_setting` distinguishes `offline` (in-laboratory) from `online` (web-based) data acquisition (Vogt et al., 2022).

3.1.3.4 Target–distractor relationship (variables 18–21). These four indicators encode the principal types of relatedness manipulated in the PWI literature. `congruency` is set to `congruent` when the distractor word is identical to the target picture name and to `incongruent` otherwise; the incongruent case corresponds to the typical PWI trial. `categorical_relation` marks distractors that share a semantic category with the target (e.g., *cat* as a distractor for the target *dog*). `associative_relation` marks distractors that are associatively but not necessarily categorically related to the target (e.g., *bone* for *dog*). `phonological_relation` marks distractors that share initial phonemes or other phonological features with the target (e.g., *dog* for *doll*). The four indicators are not mutually exclusive: a single distractor can carry several types of relatedness simultaneously, depending on the original study design.

3.1.3.5 Psycholinguistic properties of target and distractor words (variables 22–28). For both targets (variables 22–24) and distractors (variables 25–28), we provide the word length in number of letters and the Zipf frequency, a log-transformed lexical-frequency measure on a roughly 1–7 scale on which higher values indicate more frequent words (Brysbaert, n.d.; Brysbaert et al., 2018). Frequencies were computed from the deWaC corpus for German and the ukWaC corpus for English (Baroni et al., 2009); values exceeding 7 reflect a small number of very high-frequency items in these corpora. The repetition-count variables (`target_repetition_count`, `distractor_repetition_count`) encode how often a given target or distractor has previously appeared (Kurtz et al., 2018; Wöhner et al., 2022). `distractor_modality` additionally indicates whether the distractor was presented as written `text` (the canonical visual PWI variant) or as auditory `audio`, since the modality of presentation can substantially influence the time course of the interference effect.

Table 3

Overview of Dataset Variables.

#	Variable	Description	Range / Coding
Identifiers			
1	`study_id`	Study identifier	42 unique
2	`experiment_id`	Experiment identifier	86 unique
3	`participant_id`	Participant identifier	3,353 unique
Trial-level response			
4	`trial_number`	Trial number	1–900
5	`target_word`	Target word (picture name)	1,420 unique
6	`distractor_word`	Distractor word	2,529 unique
7	`rt`	Response time (ms)	150–3,000
8	`accuracy`	Response accuracy code	C / W / O
Experimental design			
9	`language`	Experimental language	EN / DE
10	`native_dutch`	Dutch native speaker	no / yes
11	`native_french`	French native speaker	no / yes
12	`has_familiarization`	Familiarization phase	no / yes
13	`soa`	Stimulus onset asynchrony	−650–300
14	`task_type`	Naming task type	covert / overt
15	`has_additional_tasks`	Additional tasks present	no / yes
16	`is_gamified`	Gamification used	no / yes
17	`collection_setting`	Collection setting	offline / online
Target–distractor relationship			
18	`congruency`	Target equals distractor	I / C
19	`categorical_relation`	Categorical relation	unrelated / related
20	`associative_relation`	Associative relation	unrelated / related
21	`phonological_relation`	Phonological relation	unrelated / related
Target properties			
22	`target_length`	Target length (letters)	2–15
23	`target_zipf_frequency`	Target Zipf frequency	2.73–9.45
24	`target_repetition_count`	Target repetition count	0–36
Distractor properties			
25	`distractor_modality`	Distractor modality	audio / text
26	`distractor_length`	Distractor length (letters)	2–17
27	`distractor_zipf_frequency`	Distractor Zipf frequency	2.73–9.56
28	`distractor_repetition_count`	Distractor repetition count	0–35

Note. Abbreviated codings in the table are expanded here. For `accuracy`: `correct` (C) marks trials in which the participant named the target picture correctly; `wrong_word` (W) marks trials in which a different word than the intended target was produced; `other_error` (O) marks any other subject-side error, such as disfluencies, omissions, or non-speech sounds. For `congruency`: `incongruent` (I) means the distractor differs from the target (the typical PWI case), while `congruent` (C) means the distractor matches the target word.

4 Technical Validation

4.1 Response Time Distributions

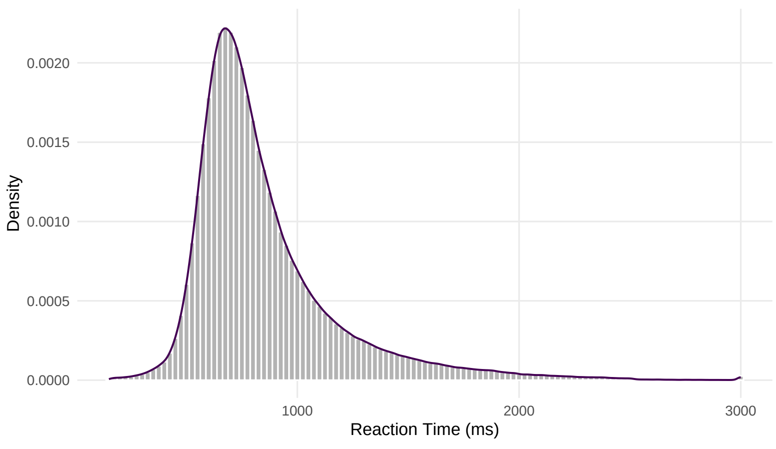
Response times across the full dataset have a right-skewed distribution typical of naming latency data, with a median of 763 ms and a mean of 853 ms ($SD = 331$ ms). The minimum observed RT is 150 ms, and the maximum is 3000 ms, reflecting the predefined cut-off values. Per-study RT distributions show variability reflecting differences in experimental design, stimulus onset asynchrony, and participant populations. Figure 3 shows the pooled RT distribution across all studies, while Figures 4 and 5 display per-study densities for the first and second half of the studies, respectively.

4.2 Trial and Participant Counts

The 42 studies contribute between 1 and 8 experiments each, with a median of 1 experiment per study; the three studies contributing the most experiments are Wei et al. (2022) (8 experiments), Sailor et al. (2009) (6 experiments), and Wöhner et al. (2022) (6 experiments). Participant counts per study range from 16 to 298 (median 54), and trial counts per study range from 1,155 to 75,697 (median 10,334). The five largest contributors by trial count account for approximately 36% of total trials, reflecting the inclusion of several large in-house and multi-experiment datasets. Figures 6 and 7 visualize these counts for each study.

Figure 3

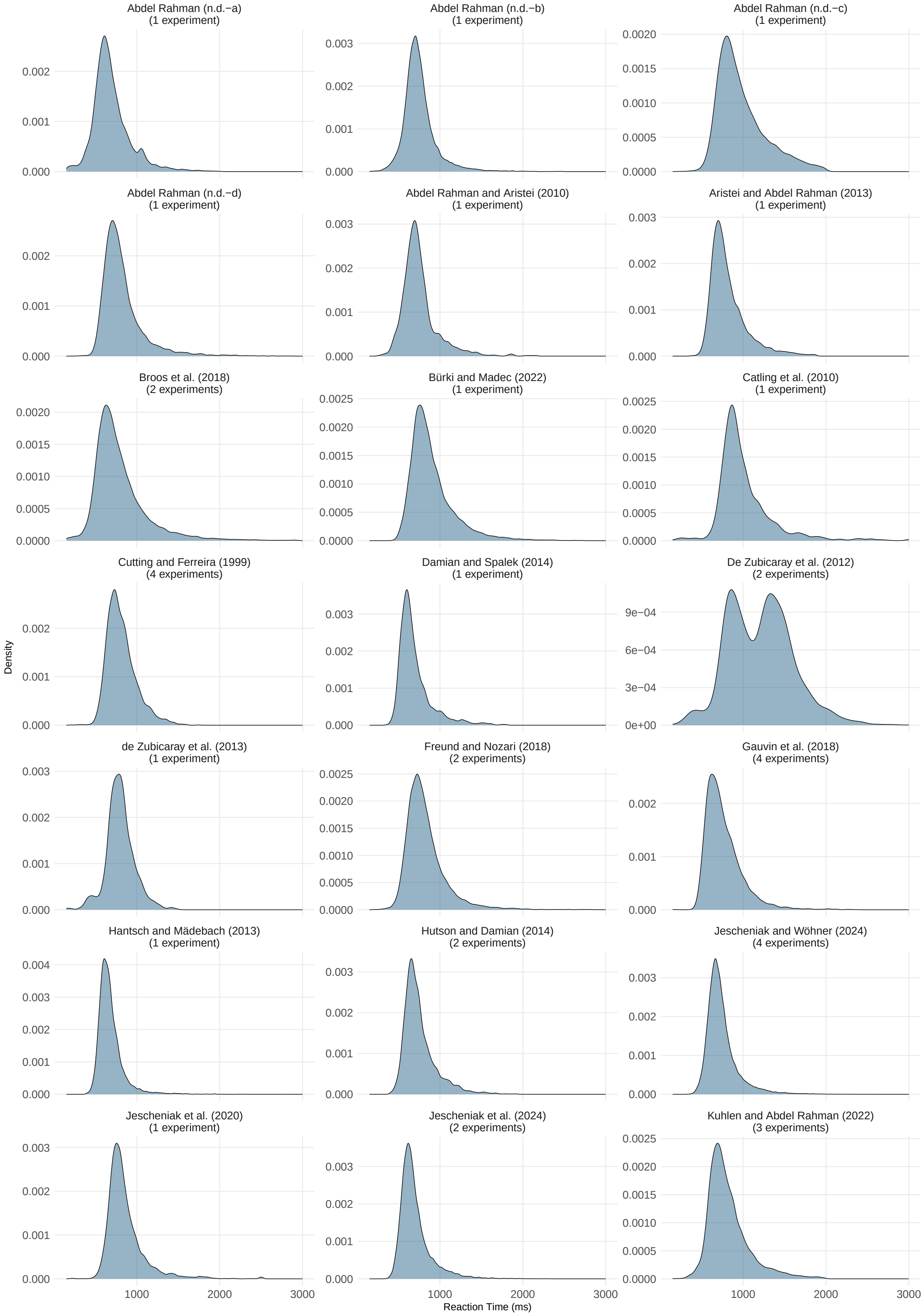
Distribution of response times across the full dataset ($n = 688,976$ trials).



Note. Histogram (grey bars) overlaid with a kernel density estimate (dark line).

Figure 4

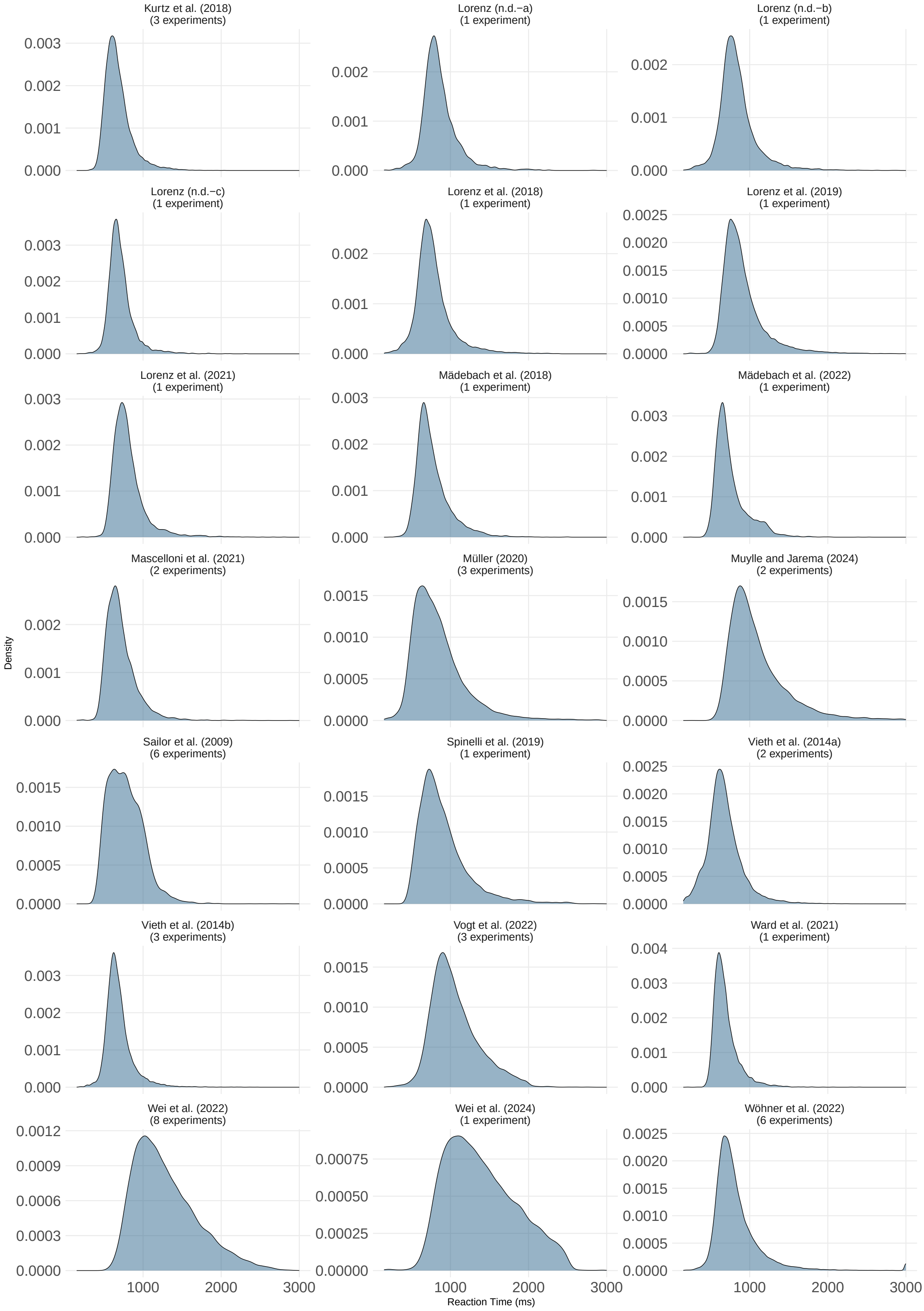
Response time distributions by study (part 1 of 2).



Note. Kernel density estimates of trial-level response times for the first 21 studies (alphabetical order).

Figure 5

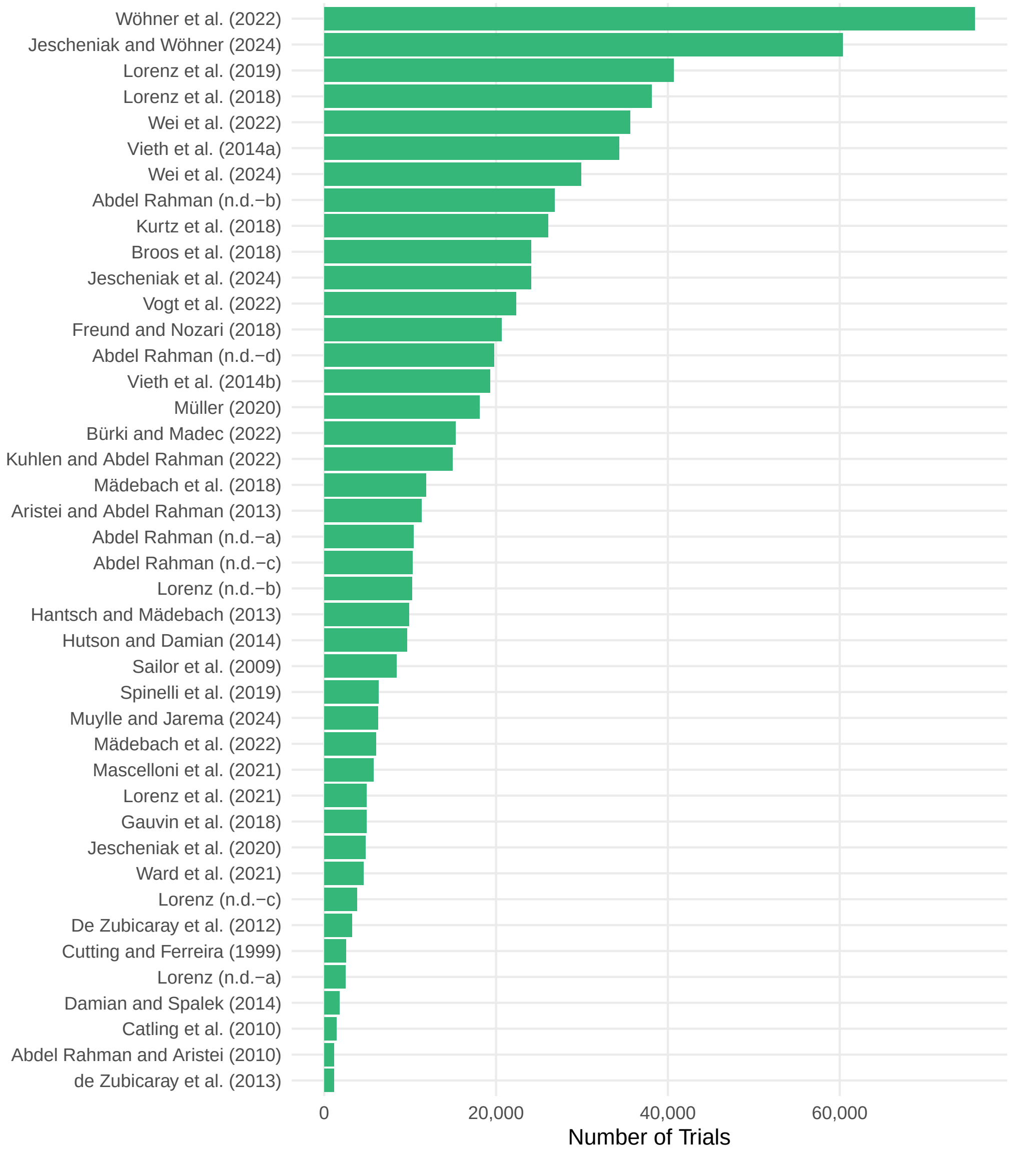
Response time distributions by study (part 2 of 2).



Note. Kernel density estimates of trial-level response times for the remaining 21 studies (alphabetical order).

Figure 6

Number of trials per study.



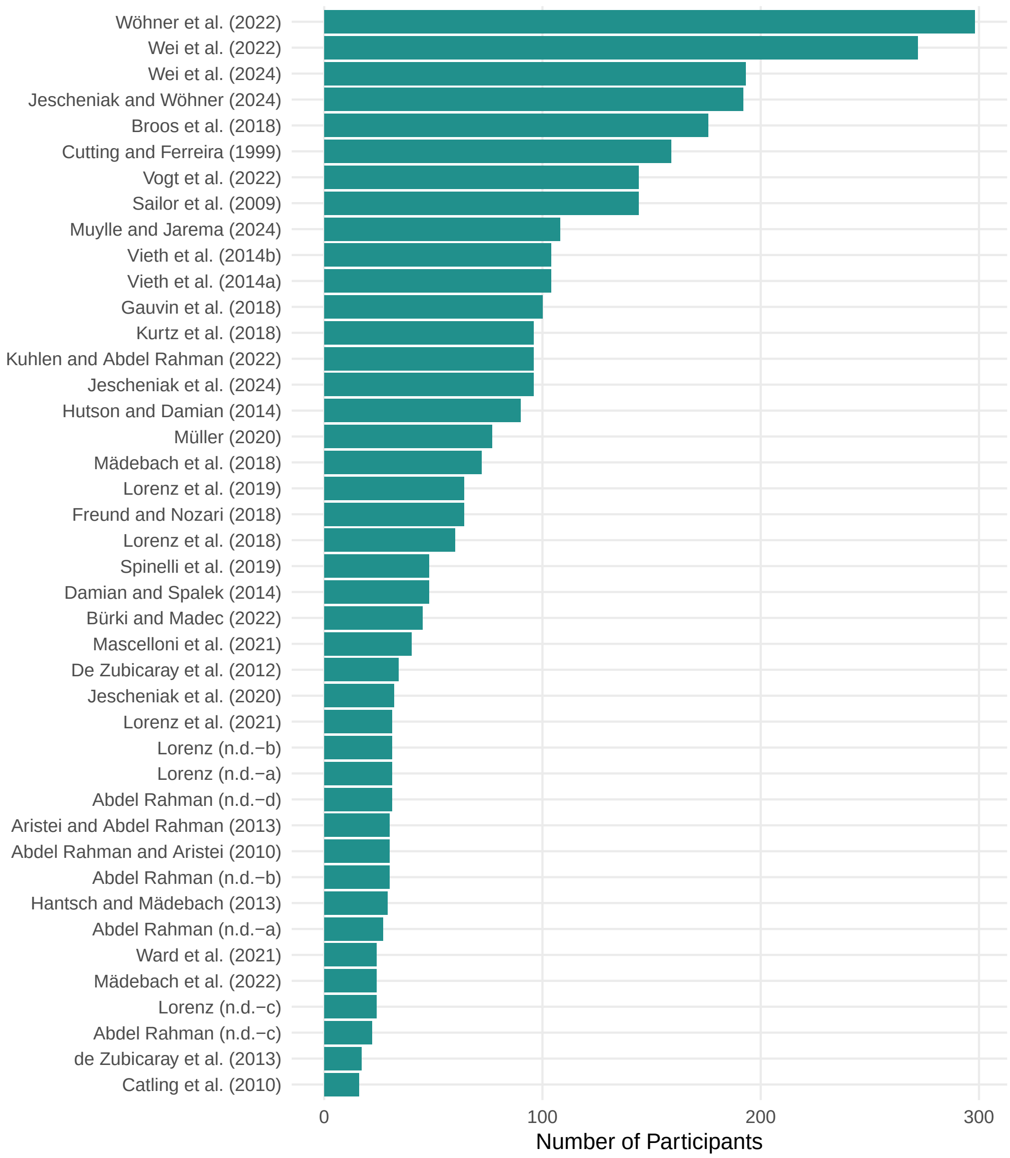
Note. Bar chart of total trial counts for each of the 42 included studies, ordered from smallest to largest.

4.3 Accuracy and Error Rates

We verified response accuracy across the dataset as part of technical validation. Of the 688,976 trials, 95.11% (655,254) are coded as `correct`, 0.67% (4,626) as `wrong_word`, 2.98% (20,566) as `other_error` (disfluencies, no-response trials, and similar subject-side errors), and 1.24% (8,530) as `NA` because the original source data did not record an accuracy code for those trials. Trials flagged by the original authors as `technical_error` (voice-key failures, microphone artefacts) were removed centrally during harmonization and are therefore absent from the final dataset.

Figure 7

Number of participants per study.



Note. Bar chart of unique participant counts for each of the 42 included studies, ordered from smallest to largest.

4.4 Missing Data

The relatedness variables (`categorical_relation`, `associative_relation`, `phono-logical_relation`) are coded as `NA` for studies that did not manipulate the corresponding dimension. The `trial_number` variable is missing for studies in which trial order was not recorded in the original data. Consequently, the repetition measures (target_repetition_count and distractor_repetition_count) are also unavailable for these studies (5.48% missing each), as their computation requires information about trial sequence. The accuracy column is missing for 1.24% of trials, as detailed in the preceding section. The proportion of missing data per variable is as follows: trial_number (5.48%), accuracy (1.24%), categorical_relation (29.82%), associative_relation (70.29%), phonological_relation (60.35%), target_repetition_count (5.48%), and distractor_repetition_count (5.48%). All remaining variables are complete and contain no missing values.

4.5 Known Limitations

Several limitations should be noted when using this dataset:

- **Language scope.** The dataset includes only English and German studies. PWI studies in other languages (e.g., Dutch, French, Mandarin) were excluded. This restriction reflects the comparatively high availability of trial-level PWI data in English and German and the fact that the authors are fluent in both languages, which was needed for stimulus and response validation; extending the dataset to additional languages is a natural direction for future releases.

- **Trial order.** For some studies, trial order was not recorded in the original data, resulting in missing values for trial_number and for the target and distractor repetition counts.

- **Accuracy data.** Twenty of the 42 studies contribute only correct trials because their raw files had already been filtered upstream; the remaining 22 studies contribute the full range of accuracy labels. Users who wish to restrict analyses to correct trials can filter on accuracy == "correct".

- **Temporal coverage.** Although no strict publication-year cutoff was applied to data inclusion, the literature search focused on studies published from 2010 onward. Earlier studies were entered through the Bürki meta-analysis or in-house channels.

- **Cross-study participant uniqueness.** The participant_id column is unique within each study but cannot be linked across studies. For the in-house corpus collected at the Neurocognitive Psychology Lab at *Humboldt-Universität zu Berlin* (i.e., `lorenz_*`, `abdel_rahman_unpub*`, `kuhlen_2022`, and `vogt_2022`), it is therefore possible that the same individual participated in more than one experiment.

5 Usage Notes

The merged dataset is distributed as a single UTF-8-encoded CSV file at `data/03_merged_data/df_merged_<date>.csv` and can be loaded in any standard analysis environment (e.g., `read.csv()` in R, `pandas.read_csv()` in Python). Each row corresponds to a single trial. The 28 variables are documented in Table 3 and fall into six logical groups: identifiers (study, experiment, participant), trial-level response (trial number, target, distractor, response time, accuracy), experimental design characteristics, target–distractor relationship indicators, target properties, and distractor properties.

- **Handling missing values.** Missingness in the dataset is structural rather than random: the relatedness indicators (`categorical_relation`, `associative_relation`, `phonological_relation`) are missing whenever a study did not manipulate that contrast, and the repetition counts together with `trial_number` are missing for studies where trial order was not recorded. We recommend filtering by the presence of the relevant variables rather than using listwise deletion across the whole dataset, which would unnecessarily discard complete rows.

- **Filtering by accuracy.** The `accuracy` column reports a harmonized response code (`correct`, `wrong_word`, `other_error`) derived from the original study-specific error labels. Technical-error trials have already been removed. For analyses restricted to correct responses, filter on `accuracy == "correct"`; for more lenient subsets, include `other_error` trials (e.g., disfluencies) as appropriate.

- **Language subsets.** Users interested in only one language can filter by the `language` variable (`EN`: English, `DE`: German). Per-language subsets preserve all variables and can be written out as separate CSV files.

- **Reproducing the dataset from source.** The full pipeline can be reproduced from raw data by running, in order, Stage1_execute_all_cleaning_scripts.R, Stage2_Read_cleaned_data_English.R, Stage2_Read_cleaned_data_German.R, and Stage3_Merge_dataset_paper.R (all located in scripts/01_data_cleaning/). Intermediate outputs are written to data/02_cleaned_data/ and the final merged file to data/03_merged_data/.

- **Extending the dataset.** The full pipeline—raw data, study-specific cleaning scripts, harmonization code, and the merged file—is deposited as a self-contained project on OSF (https://doi.org/10.17605/OSF.IO/2B3SX). Researchers who wish to extend the dataset with additional PWI studies can download the entire project, place raw data for the new study in data/01_single_studies/<study_name>/Data/, add a clean_<study_name>.R script in the corresponding Scripts/ folder that produces output matching the 28-variable schema, update data/overview_studies.xlsx, and re-run Stage1_execute_all_cleaning_scripts.R followed by the language-specific loaders and Stage3_Merge_dataset_paper.R. The result is a locally extended copy of the merged dataset that follows the same harmonized schema as the published version.

6 Data Availability

The full dataset, together with all raw study files, study-specific cleaning scripts, harmonization code, and the final merged CSV, is openly available on the Open Science Framework (OSF) at https://doi.org/10.17605/OSF.IO/2B3SX. All materials are released under a Creative Commons Attribution 4.0 International (CC-BY 4.0) license. Of the 42 included studies, eight had already released their trial-level data under CC-BY 4.0 before inclusion; for the remaining 34 studies, written permission to redistribute trial-level data under the same license was obtained from the corresponding authors. No access restrictions apply.

7 Code Availability

All custom code used to generate the dataset is released under a CC-BY 4.0 license and is openly available alongside the data on OSF (https://doi.org/10.17605/OSF.IO/2B3SX) in the `scripts/` directory. The pipeline consists of study-specific cleaning scripts (`data/01_single_studies/<study>/Scripts/clean_<study>.R`) and four orchestrating scripts in `scripts/01_data_cleaning/`: `Stage1_execute_all_cleaning_scripts.R`, `Stage2_Read_cleaned_data_English.R`, `Stage2_Read_cleaned_data_German.R`, and `Stage3_Merge_dataset_paper.R`. All analyses were performed in R (version 4.3.0). To reproduce the merged dataset, execute the four stage scripts in the order listed above.

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Author Contributions

L.S. conceived the study, performed the data collection, cleaning, harmonization, and analysis, and drafted the manuscript. R.A.R. contributed to study design and methodology, provided data, supervised the project, and contributed to writing (review & editing). V.G. contributed to data collection and preprocessing, developed analysis scripts, and contributed to writing (review & editing). A.B., and A.L., provided data, contributed to methodology, and contributed to writing (review & editing). K.S. contributed to methodology, and contributed to writing (review & editing). F.G. contributed to study design and methodology, supervised the project, and contributed to writing (review & editing).

Competing Interests

The authors declare no competing interests.

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